

Hello-World Final Presentation

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Today's Agenda:

1. Problem Statement
2. Problem Solving
3. Challenges
4. Open Day
5. Feedback

Problem Statement

01

“Delivery drivers face difficulties accessing restrooms while working, which undermines their health and professional dignity.”

Problem Solving

02

1. Google Map like navigation
2. New York open data source image
3. Mapbox as map

Facility	Location	Owner	Status	Open	Hours of ...	Access
Barry Park II	Manh...	NYC Parks	Not Operational			
Carroll Park	Manh...	NYC Parks	Not Operational			NYC Parks
Capen Park	Manh...	NYC Parks	Not Operational			
Clinton Park	Manh...	NYC Parks	Operational	Year Round	Apr - Sept	
Convent Ave	Manh...	NYC	Closed for Ren	Year Round		NYC Parks
Agard Library	Manh...	NYC	Closed for Ren	Public		
Deerpark	Manh...	NYC Parks	Operational	Year Round	Year-Round	
Madison Park	Manh...	NYC Parks	Not Operational			NYC Parks
Greenwich	Manh...	NYC	Operational	Year Round	Monday - 10 pm	NYC Parks

```
import { useEffect, useRef, useState } from 'react';
import mapboxgl from 'mapbox-gl';
import MapboxDirections from '@mapbox/mapbox-gl-directions/dist/mapbox-gl-directions';
import '@mapbox/mapbox-gl-directions/dist/mapbox-gl-directions.css';
import './MapPage.css';

mapboxgl.accessToken = 'pk.eyJ1Ijoic2FuZnJhbnR1c2NvLW4wIiwiaS1Y3IGInWtaTg8aDlyYzA3enky';

function MapPage() {
  const mapContainer = useRef(null);
  const map = useRef(null);
  const directionsControl = useRef(null);
  const bathroomMarkersRef = useRef([]);
  const restaurantMarkersRef = useRef([]);
  const fetchTimeoutRef = useRef(null);

  const [bathrooms, setBathrooms] = useState([]);
  const [restaurants, setRestaurants] = useState([]);
  const [userLocation, setUserLocation] = useState([173.9853, 48.7580]);
  const [selectedBathroom, setSelectedBathroom] = useState(null);
  const [selectedRestaurant, setSelectedRestaurant] = useState(null);
  const [isLoading, setIsLoading] = useState(true);
  const [showDirections, setShowDirections] = useState(false);
  const [currentZoom, setCurrentZoom] = useState(13);

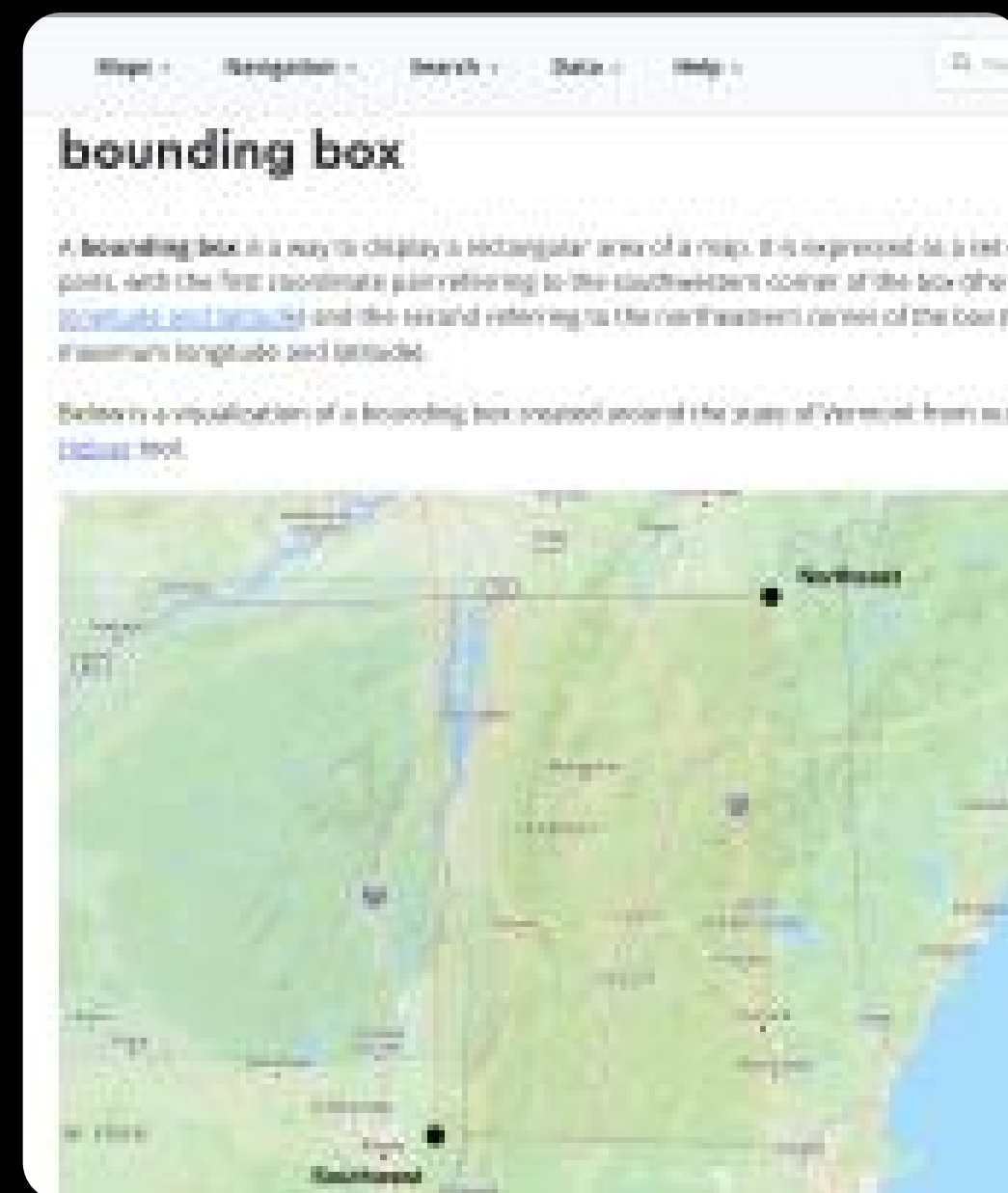
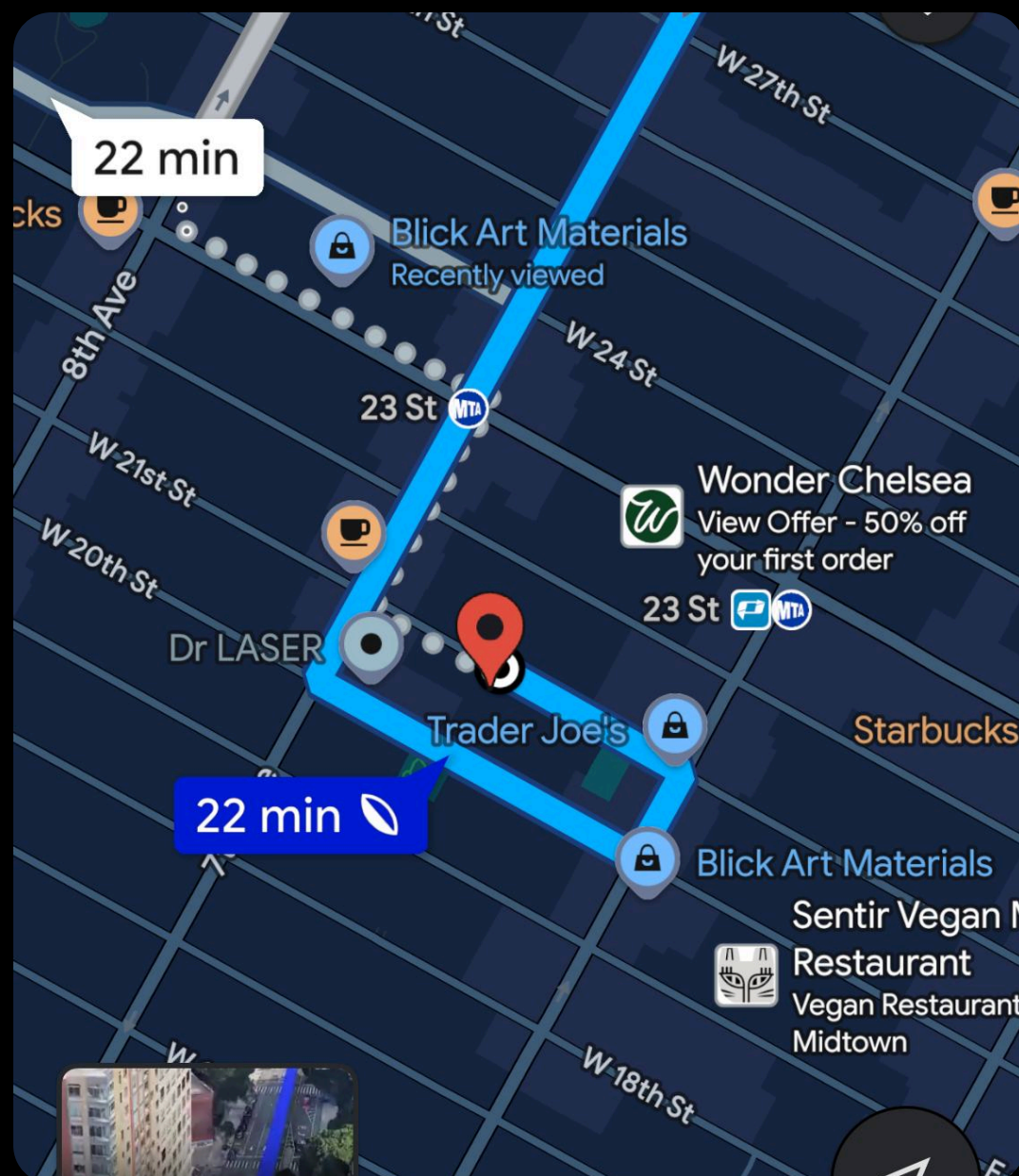
  const MIN_ZOOM_FOR_MARKERS = 14;
```

```
// 지도 초기화
useEffect(() => {
  if (map.current) return;

  console.log('Initializing Mapbox...');

  try {
    const nycBounds = [
      [-74.3, 40.4],
      [-73.6, 41.0]
    ];

    map.current = new mapboxgl.Map({
      container: mapContainer.current,
      style: 'mapbox://styles/mapbox/standard',
      center: userLocation,
      zoom: 13,
      attributionControl: false,
      maxBounds: nycBounds
    });
  }
});
```



```
// 레스토랑 데이터
async function fetchRestaurantsInBounds(sw, se) {
  try {
    const query = `https://data.cityofnewyork.us/resource/43in-pn8.json`;

    const response = await fetch(query);
    const data = await response.json();

    console.log(`Loaded ${data.length} restaurants`);

    const valid = data.filter(r => r.latitude && r.longitude);
    const filtered = filterByDistance(valid, 8.001);

    setRestaurants(filtered);

    restaurantMarkersRef.current.forEach(m => m.remove());
    restaurantMarkersRef.current = [];

    addRestaurantMarkers(filtered);
  } catch (error) {
    console.error('❌ Fetch restaurants error:', error);
  }
}
```

```

ync function fetchBathroomsInBounds(sw, ne) {
  try {
    const query = `https://data.cityofnewyork.us/resource/17`;

    const response = await fetch(query);
    const data = await response.json();

    console.log(`Loaded ${data.length} bathrooms`);

    const valid = data.filter(b => b.latitude && b.longitude);
    const filtered = filterByDistance(valid, 0.001);

    setBathrooms(filtered);

    bathroomMarkersRef.current.forEach(m => m.remove());
    bathroomMarkersRef.current = [];

    addBathroomMarkers(filtered);
  } catch (error) {
    console.error(`❌ Fetch bathrooms error:`, error);
  }
}

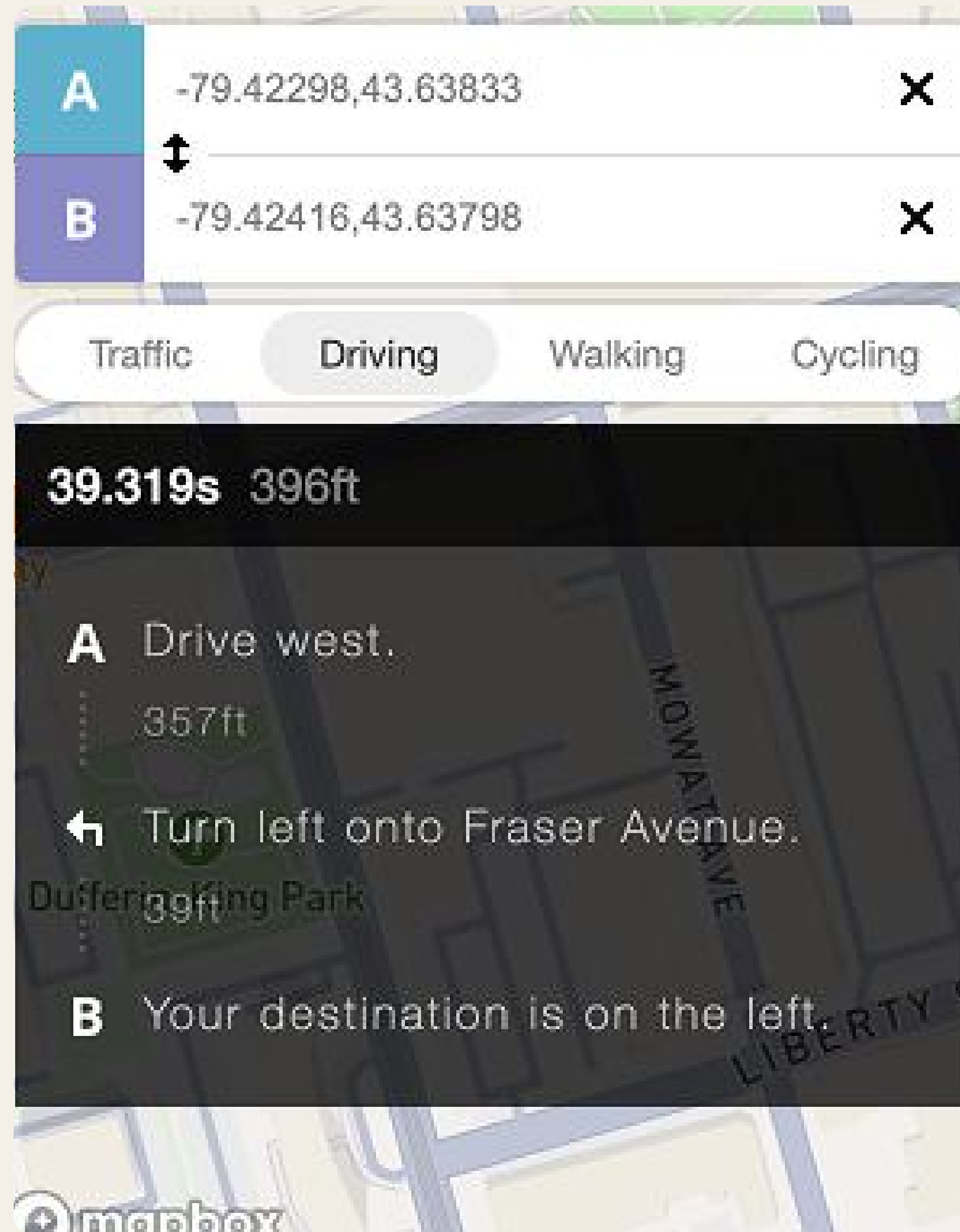
```

Challenges

03

Challenges

1. Mapbox's CSS complicit (when making glass morphic style)
2. Mobile and web difference (CSS)
3. Speed to load data



Overcome

- 1. Mapbox's CSS complicit (when making glass morphic style)
- 2. Mobile and web difference (CSS)
- 3. Speed to load data

```
/* ===== */
/* Mapbox Controls ~ Glassmorphic (최종 수정) */
/* ===== */

/* 컨트롤 그룹 전체 */
.mapboxgl-ctrl-group {
  background: linear-gradient(to top right, transparent 49%, #000 49% 51%, #000 51% 53%, transparent 53%);
  backdrop-filter: blur(20px) saturate(180%) !important;
  -webkit-backdrop-filter: blur(20px) saturate(180%) !important;
  border: 1px solid #000;
  border-radius: 12px !important;
  box-shadow:
    0 0px 32px #000,
    inset 0 1px 0 #000;
  overflow: hidden !important;
}

/* 개별 버튼 */
.mapboxgl-ctrl-group button {
  background: transparent !important;
  background-color: transparent !important;
  border: none !important;
  width: 36px !important;
  height: 36px !important;
  transition: all 0.2s ease !important;
  display: flex !important;
  align-items: center !important;
  justify-content: center !important;
}
```

```
useEffect(() => {
  // 중요한 리소스 프리로드 -> 속도이슈 문제
  const preloadImages = [
    '/toilet.png',
    '/restaurants.png',
    '/loading-toilet.gif'
  ];

  const imagePromises = preloadImages.map(src => {
    return new Promise((resolve) => {
      const img = new Image();
      img.onload = resolve;
      img.onerror = resolve; // 에러나도 계속 진행
      img.src = src;
    });
  });
});
```

```
// =====
// 최적화 1: 디폴트 줌
// =====

map.current.on('moveend', () => {
  if (fetchTimeoutRef.current) clearTimeout(fetchTimeoutRef.current);
  fetchTimeoutRef.current = setTimeout(() => {
    const zoom = map.current.getZoom();
    if (zoom >= MIN_ZOOM_FOR_MARKERS) fetchDataInBounds();
  }, 800); // 500ms -> 800ms (덜 자주 fetch)
```

```
// =====
// 최적화 2: 마커 제한
// =====

function filterByDistance(items, minDistance) {
  const filtered = [];
  const maxMarkers = 100; // 최대 마커 수 제한 추가

  items.forEach(item => {
    if (filtered.length >= maxMarkers) return; // 100개 넘으면 중단

    const tooClose = filtered.some(existing) => {
      const latDiff = Math.abs(parseFloat(item.latitude) - parseFloat(existing.latitude));
      const lngDiff = Math.abs(parseFloat(item.longitude) - parseFloat(existing.longitude));
      return latDiff < minDistance && lngDiff < minDistance;
    };

    if (!tooClose) filtered.push(item);
  });

  return filtered;
}
```

```
// 최적화 3: Fetch 범위와 개수
// =====
async function fetchDataInBounds() {
  if (!map.current) return;

  setIsLoading(true);

  try {
    const bounds = map.current.getBounds();
    const sw = bounds.getSouthWest();
    const ne = bounds.getNorthEast();

    // Promise.allSettled로 변경 (무시 실패해도 다른 건 실행)
    const results = await Promise.allSettled([
      fetchBathroomsInBounds(sw, ne),
      fetchRestaurantsInBounds(sw, ne)
    ]);

    results.forEach((result, index) => {
      if (result.status === 'rejected') {
        console.warn('Fetch ${index === 0 ? 'bathrooms' : 'restaurants'} failed: ', result.reason);
      }
    });
  } finally {
    setIsLoading(false);
  }
}
```

```
// =====
async function fetchDataInBounds() {
  if (!map.current) return;

  // 초기 로딩 시 setisLoading 사용
  // 이후 fetch 시 setisLoading 사용
  const isInitialLoad = !bathrooms.length && !restaurants.length;

  if (isInitialLoad) {
    setIsLoading(true);
  } else {
    setIsFetching(true); // 로딩 바 대신 fetch
  }

  try {
    const bounds = map.current.getBounds();
    const sw = bounds.getSouthWest();
    const ne = bounds.getNorthEast();
  }
}
```

Open Day

04

Here's our TD app on mobile

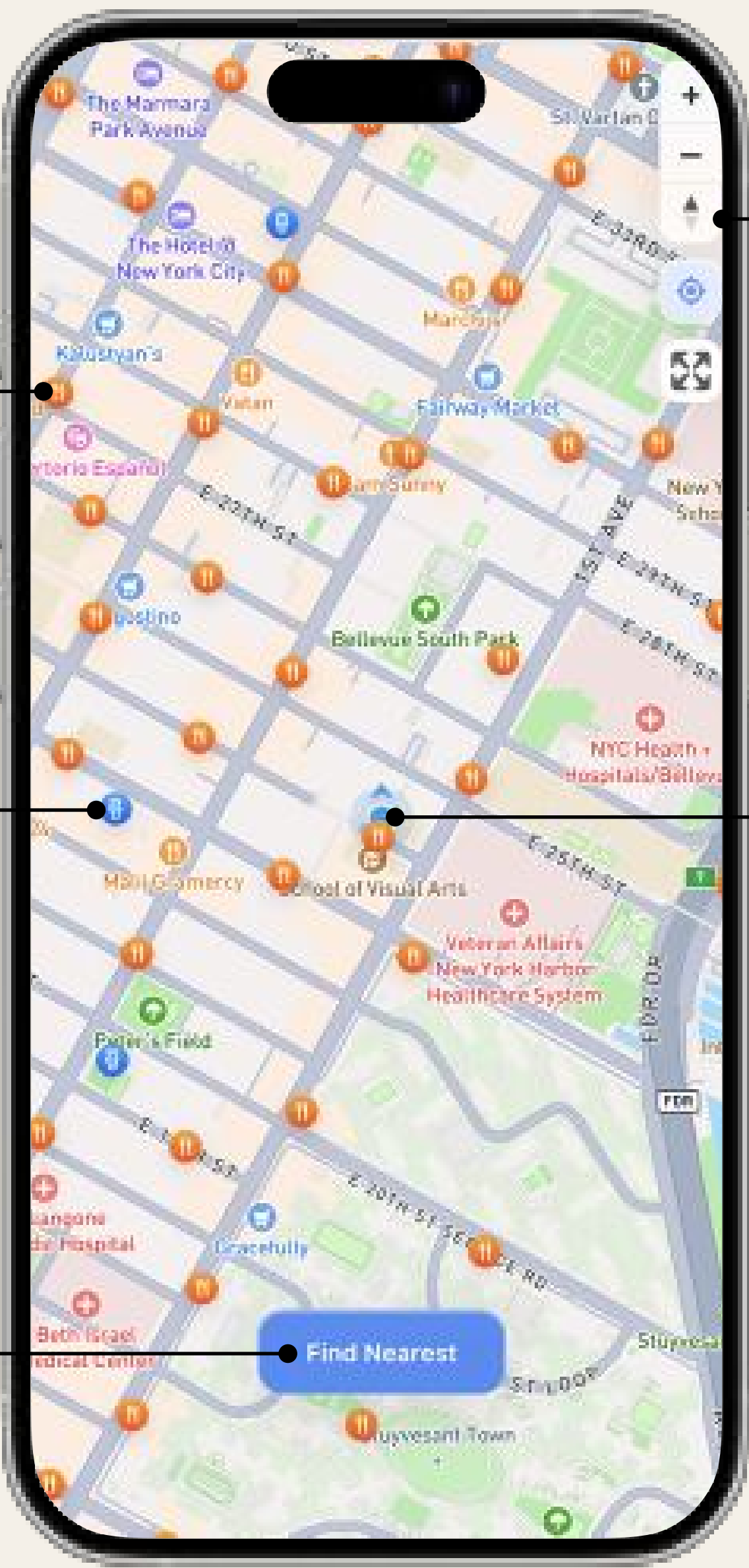
TD Web App Page

- 1. Search : <https://hello-world-2025-final.vercel.app/>
- 2. Allow location
- 3. Search **Toilet** you need!

EXAMPLE OF TOILET WITHIN RESTAURANT

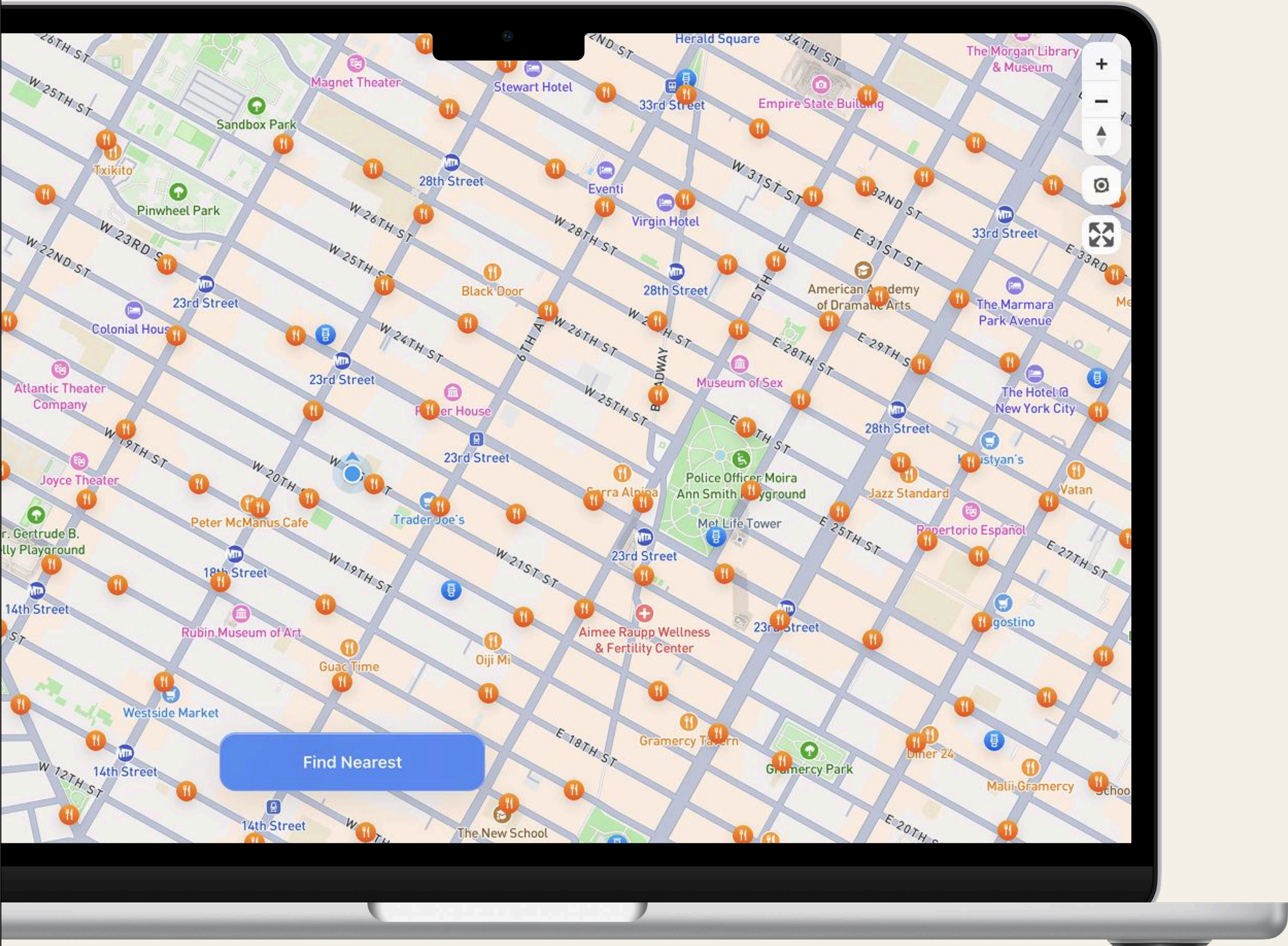
EXAMPLE OF TOILET ONLY PLACES

FIND NEAREST TOILET AROUND LOCATION



MAP VIEWING CONTROL PANEL (ZOOM IN/OUT, CURRENT LOCATION, VIEW WHOLE MAP)

CURRENT LOCATION



Here’s ourTD
app on laptop

TD Web App Page

1. Search : <https://hello-world-2025-final.vercel.app/>
2. Allow location
3. Search **Toilet** you need!



Feedback

05

Main Feedbacks



Commute Options

Walking/ Bike/ PublicTransportation &
More options for people to choose



Locating Parking

Adding parking places' options that
goes with the toilet around



Redirection

Being able to jump back from the map
page to the original front page



Loading Time

Enhance the loading time for the
map/data



Time of the day

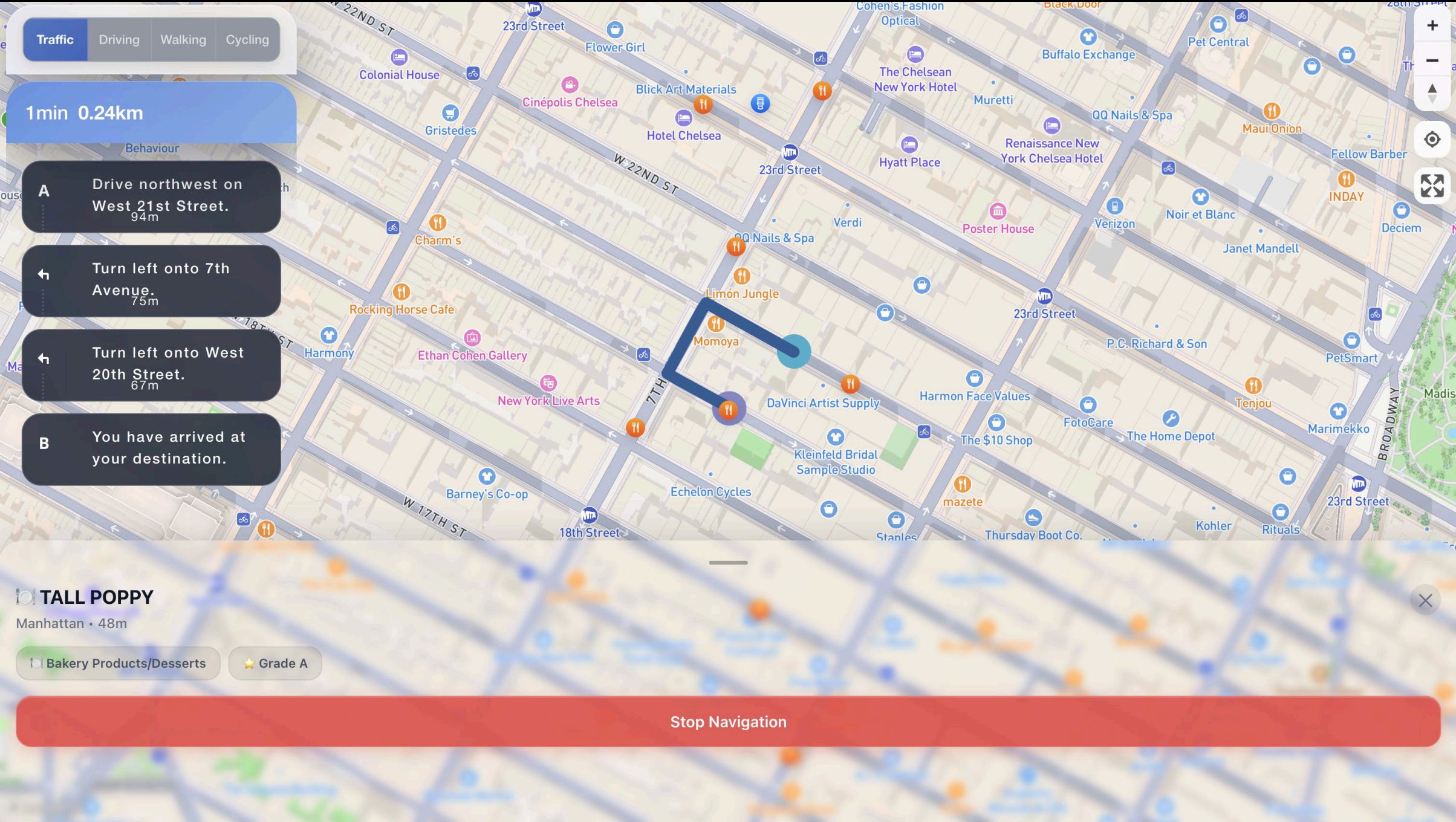
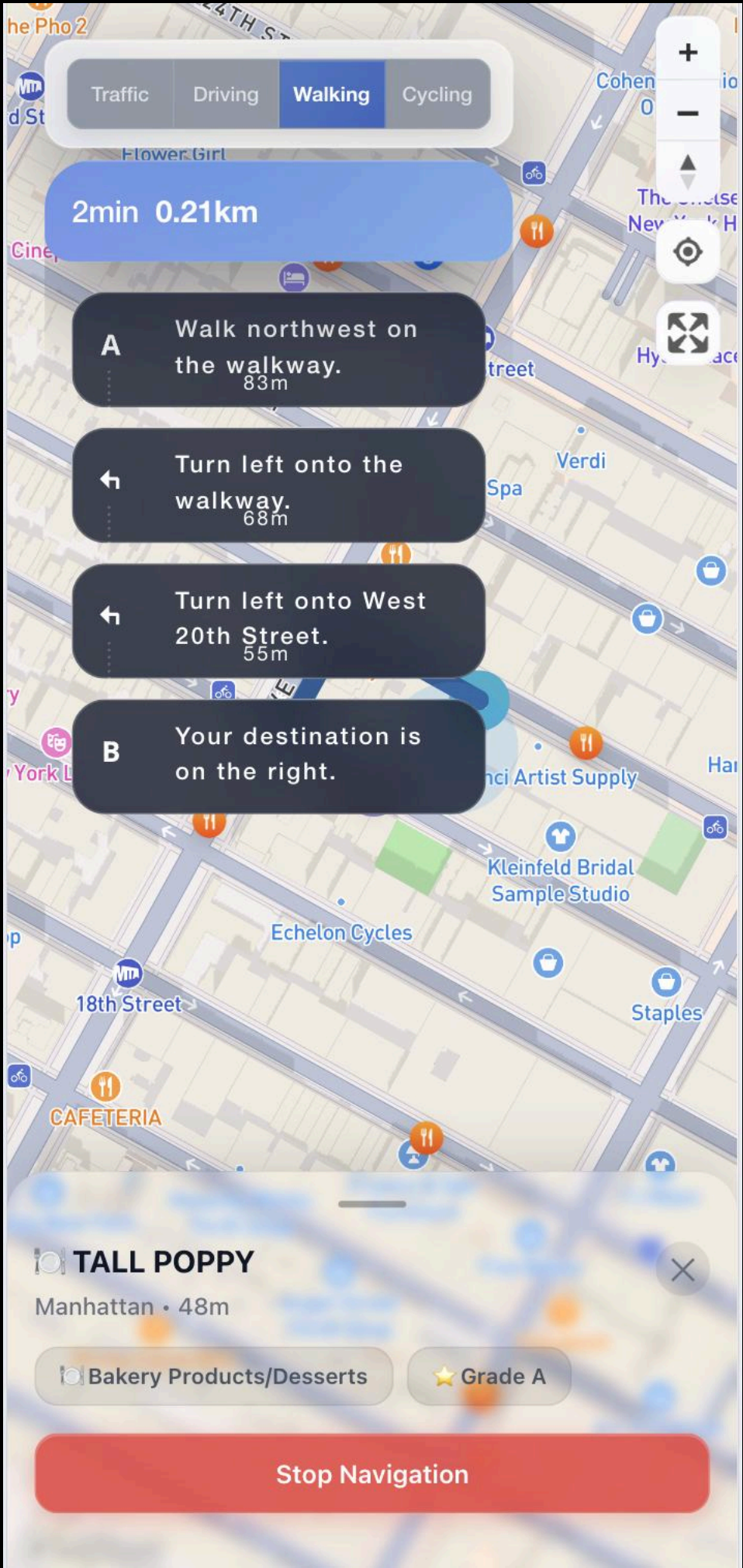
Since some toilet spots are in the
restaurant, it is better to include the
opening time of those places



Rating System

Would be better if the user are able to
doo rating too as well, so it is not a one
side information provide

```
{
  vercel.json > [ ] rewrites > { } 0
  1 {
  2   "rewrites": [
  3     {
  4       "source": "/api/:path*",
  5       "destination": "/api/:path*"
  6     },
  7     {
  8       "source": "/:path*",
  9       "destination": "/index.html"
 10     }
 11   ],
 12   "headers": [
 13     {
 14       "source": "/*",
 15       "headers": [
 16         {
 17           "key": "Cache-Control",
 18           "value": "public, max-age=0, must-revalidate"
 19         }
 20       ]
 21     }
 22   ]
 23 }
```



Thank you!

Reach out to us if you have questions.
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